

EV POWERTRAIN SIMULATION

USING MATLAB



PROJECT REPORT

ABSTRACT

This project presents the modeling and simulation of an Electric Vehicle (EV) Powertrain using MATLAB. The objective of the project is to understand vehicle longitudinal dynamics, motor performance, battery energy consumption, and regenerative braking behavior through a physics-based simulation model.

The developed model calculates driving force, rolling resistance, aerodynamic drag, vehicle acceleration, motor torque, power consumption, and battery State of Charge (SOC) over time. A dynamic throttle input is used to simulate realistic driving conditions. The project also incorporates regenerative braking, allowing energy recovery during deceleration and improving battery efficiency.

MATLAB is used to implement numerical calculations and visualize system performance through multiple plots such as velocity, torque, power, and SOC variation. The simulation provides insight into EV energy flow and demonstrates how different forces affect vehicle motion and battery behavior.

This project serves as a simplified EV powertrain model for educational purposes and helps build understanding of electric vehicle operation, vehicle dynamics, and MATLAB-based engineering simulation techniques.

INTRODUCTION

Electric Vehicles (EVs) are becoming an important alternative to conventional internal combustion engine vehicles due to their higher efficiency, lower emissions, and reduced dependence on fossil fuels. Unlike traditional vehicles that use fuel combustion to generate motion, EVs use electrical energy stored in batteries to power electric motors.

An EV powertrain is the system responsible for converting electrical energy into mechanical motion. It consists of key components such as the battery pack, motor controller, electric motor, transmission system, and wheels. These components work together to provide vehicle propulsion and control energy usage.

Understanding EV powertrain behaviour is essential for analysing vehicle performance, battery efficiency, and energy management. Vehicle motion depends on several physical forces including driving force, rolling resistance, and aerodynamic drag. These forces influence acceleration, velocity, torque demand, and power consumption.

In this project, a simplified EV powertrain model is developed in MATLAB to simulate longitudinal vehicle dynamics. The model calculates vehicle speed, motor torque, power demand, and battery State of Charge (SOC) over time. Regenerative braking is also included to represent energy recovery during braking conditions.

The purpose of this simulation is to provide a clear understanding of how EV systems operate and how mathematical models can be used to predict vehicle behaviour under dynamic driving conditions. MATLAB is used as the simulation platform due to its strong numerical computation and visualization capabilities.

OBJECTIVES

The main objectives of this project are:

- To develop a MATLAB-based Electric Vehicle (EV) powertrain simulation model
- To understand the longitudinal dynamics of an electric vehicle
- To calculate driving force, resistive forces, and net vehicle force
- To simulate vehicle acceleration and velocity response over time
- To estimate motor torque and power consumption during vehicle operation
- To monitor battery State of Charge (SOC) variation during driving conditions
- To implement regenerative braking for energy recovery during deceleration
- To visualize EV performance using MATLAB plots
- To understand the relationship between vehicle motion and battery energy usage
- To gain practical experience in MATLAB-based engineering simulation and EV modelling.

PARAMETERS USED

PARAMETER	VALUE	UNIT
Vehicle Mass	1200	Kg
Gravity	9.81	m/s ²
Rolling Resistance Coefficient	0.01	--
Air Density	1.2	Kg/m ³
Frontal Area	2.2	m ²
Drag Coefficient	0.3	--
Wheel Radius	0.3	m
Battery Capacity	50	Ah
Time Step	0.1	sec
Simulation Time	3500	sec

EV POWERTRAIN STRUCTURE



MATHEMATICAL MODEL

The EV powertrain simulation is based on vehicle longitudinal dynamics and battery energy equations. The mathematical model calculates the forces acting on the vehicle, updates velocity, estimates motor torque and power, and tracks battery State of Charge (SOC).

1. Driving Force

The driving force is generated by the electric motor to move the vehicle forward.

$$F_{drive} = m \times a$$

Where:

- F_{drive} = Driving force (N)
- m = Vehicle mass (kg)
- a = Vehicle acceleration (m/s²)

2. Rolling Resistance Force

Rolling resistance is caused by tyre deformation and road friction.

$$F_{roll} = m \times g \times C_r$$

Where:

- F_{roll} = Rolling resistance force (N)
- g = Gravitational acceleration (m/s²)
- C_r = Rolling resistance coefficient

3. Aerodynamic Drag Force

Aerodynamic drag opposes vehicle motion due to air resistance.

$$F_{drag} = \frac{1}{2} \rho A C_d v^2$$

Where:

- F_{drag} = Aerodynamic drag force (N)
- ρ = Air density (kg/m³)
- A = Frontal area (m²)
- C_d = Drag coefficient
- v = Vehicle velocity (m/s)

4. Net Force Acting on Vehicle

The total force acting on the vehicle is obtained by subtracting resistive forces from driving force.

$$F_{total} = F_{drive} - F_{roll} - F_{drag}$$

Where:

- F_{total} = Net vehicle force (N)

5. Vehicle Acceleration

Using Newton's Second Law:

$$a_{actual} = \frac{F_{total}}{m}$$

Where:

- a_{actual} = Actual acceleration (m/s²)

6. Velocity Update Equation

Vehicle velocity is updated using Euler numerical integration.

$$v_{new} = v_{old} + a_{actual} \times dt$$

Where:

- v_{new} = Updated velocity (m/s)
- dt = Simulation time step (s)

7. Motor Torque Calculation

Motor torque is calculated using wheel radius.

$$Torque = F_{drive} \times R$$

Where:

- Torque = Motor torque (Nm)
- R = Wheel radius (m)

8. Motor Power Calculation

Power required by the motor is:

$$Power = F_{drive} \times v$$

Where:

- Power = Motor power (W)

9. Battery SOC update

Battery State of Charge decreases as energy is consumed.

$$SOC_{i+1} = SOC_{(i)} - \frac{Power_i \times dt}{Capacity \times 3600}$$

Where:

- SOC = Battery State of Charge (%)
- Capacity = Battery capacity (Ah)
- 3600 = Conversion factor from hours to seconds.

10. Regenerative Braking Model

During braking, motor behaves as a generator.

Condition:

$$F_{drive} < 0$$

Regenerative power:

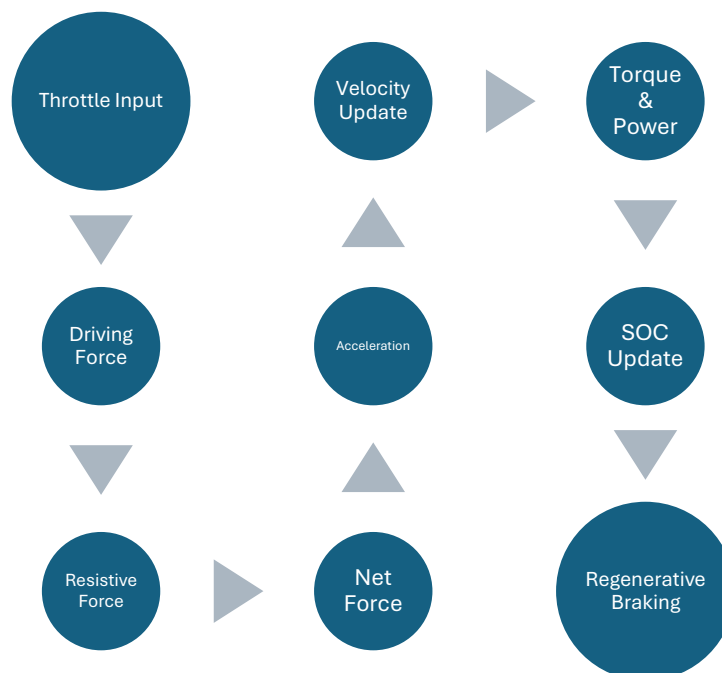
$$Power_{regen} = \eta_{regen} \times F_{drive} \times v$$

Where:

- η_{regen} = Regenerative efficiency (0.65)

This recovered energy is added back to the battery SOC.

SUMMARY OF MODEL FLOW



MATLAB IMPLEMENTATION

The EV Powertrain simulation is implemented in MATLAB using a time-domain numerical model. The program calculates vehicle motion and battery behaviour step-by-step for each simulation time interval.

The implementation is divided into several stages.

1. Initialization of Parameters

All physical parameters of the vehicle and battery are defined.

```
m = 1200;    % Vehicle mass
g = 9.81;    % Gravity
Cr = 0.01;   % Rolling resistance coefficient
rho = 1.2;   % Air density
Cd = 0.3;    % Drag coefficient
A = 2.2;     % Frontal area
R = 0.3;     % Wheel radius
Capacity = 50; % Battery capacity
```

These values remain constant during simulation.

2. Time Definition

Simulation time and step size are defined.

```
dt = 0.1;
t = 0: dt: 3500;
```

Where:

- *dt* = simulation step size
- *t* = total simulation time vector

3. Array Pre-allocation

Arrays are initialized before the loop to improve MATLAB execution speed.

```
velocity = zeros(1, length(t));
Torque = zeros(1, length(t));
Power = zeros(1, length(t));
SOC = zeros(1, length(t));
```

Pre-allocation prevents dynamic memory resizing.

4. Initial Condition

Initial conditions are assigned.

$$\begin{aligned}v &= 0; \\SOC(1) &= 100;\end{aligned}$$

Meaning:

- Vehicle starts from rest
- Battery starts fully charged

5. Main Simulation Loop

A for loop is used to simulate vehicle behaviour over time.

$$\text{for } i = 1:\text{length}(t)$$

Each iteration represents one time step.

6. Throttle Input

Throttle is used to generate acceleration demand.

$$\text{throttle} = 0.5 + 0.5 * \sin(0.3 * t(i));$$

This creates varying acceleration.

7. Force Calculation

Driving force and resistance forces are computed.

$$\begin{aligned}F_{drive}(i) &= m * a(i); \\F_{roll} &= m * g * Cr; \\F_{drag}(i) &= \frac{1}{2} \rho * A * Cd * v^2;\end{aligned}$$

8. Net Force Calculation

Total force acting on vehicle:

$$F_{total}(i) = F_{drive}(i) - F_{roll} - F_{drag}(i)$$

9. Vehicle Dynamics Update

Acceleration and velocity are updated.

$$\begin{aligned}a_{Actual}(i) &= \frac{F_{total}(i)}{m}; \\v &= v + a_{Actual}(i) * dt; \\velocity(i) &= v;\end{aligned}$$

10. Torque Calculation

Motor Torque is calculated.

$$Torque(i) = F_{drive}(i) * R;$$

11. Power Calculation

Motor power is estimated.

$$Power(i) = F_{drive}(i) * v;$$

12. Regenerative Braking

Regenerative braking activates when force becomes negative.

```

if F_drive(i) >= 0
    Power(i) = F_drive(i) * v;
elseif F_drive(i) < 0
    Power(i) = 0.65 * F_drive(i) * v;
end

```

13. SOC Update

Battery SOC updates at every time step.

$$SOC(i+1) = SOC(i) - \frac{Power(i) * dt}{Capacity * 3600}$$

14. Plot Generation

MATLAB 'tiledlayout' is used for visualization.

```
tiledlayout(6,2)
```

Generated plots include:

- Speed vs Time
- SOC vs Distance
- SOC vs Time
- Distance vs Time
- Torque vs Time
- Power vs Time
- Current vs Time
- Voltage vs Time
- RPM vs Time

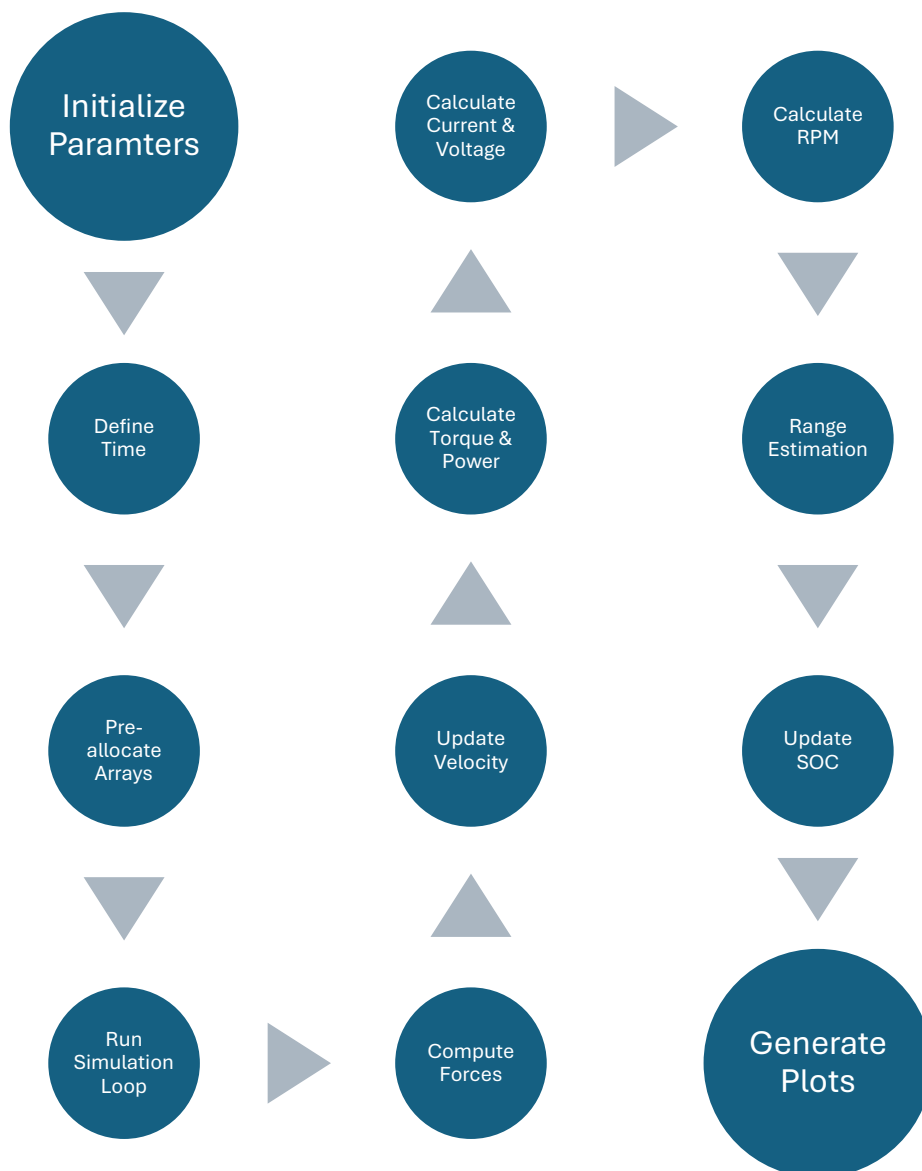
15. Logical Indexing for Regenerative Region

Logical indexing highlights regen behaviour.

$$drive_idx = Power \geq 0;$$
$$regen_idx = Power < 0;$$

Used to colour power graph regions separately.

SUMMARY OF MATLAB WINDOW



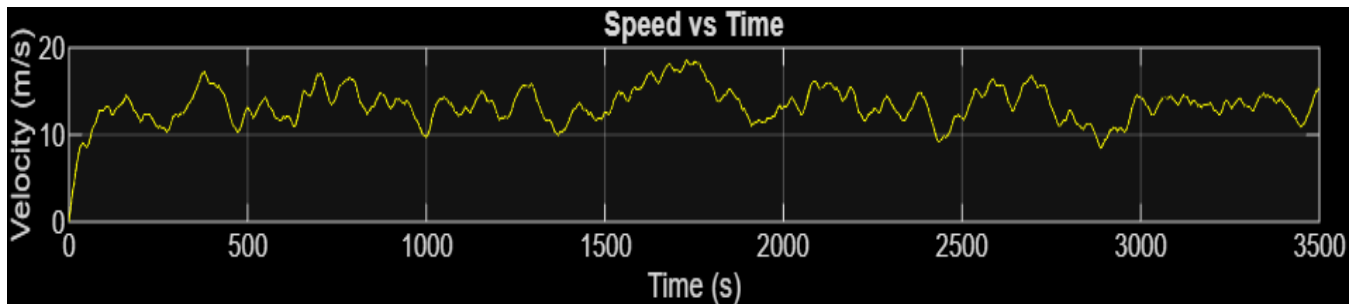
RESULTS AND GRAPHS

The MATLAB simulation generates multiple plots to evaluate the performance of the EV powertrain model under dynamic driving conditions.

These graphs help analyse vehicle motion, battery behaviour, motor performance, and energy consumption.

1. Speed vs Time

This graph shows vehicle velocity variation over simulation time.



Observation:

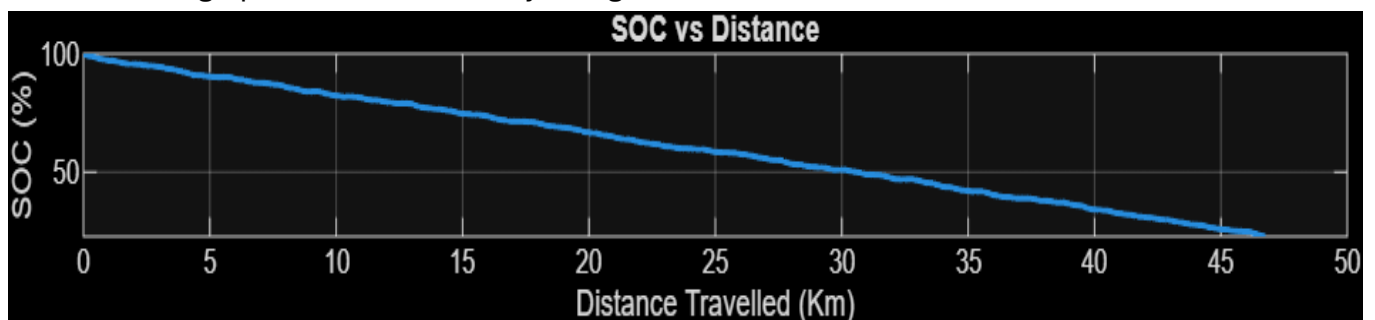
- Speed increases during acceleration
- Speed reduces during braking
- Dynamic throttle causes speed fluctuation
- Vehicle reaches steady-state speed at certain intervals

Purpose:

- Analyse vehicle dynamic response
- Observe acceleration behaviour

2. SOC vs Distance

This graph shows how battery charge decreases with travelled distance.



Observation:

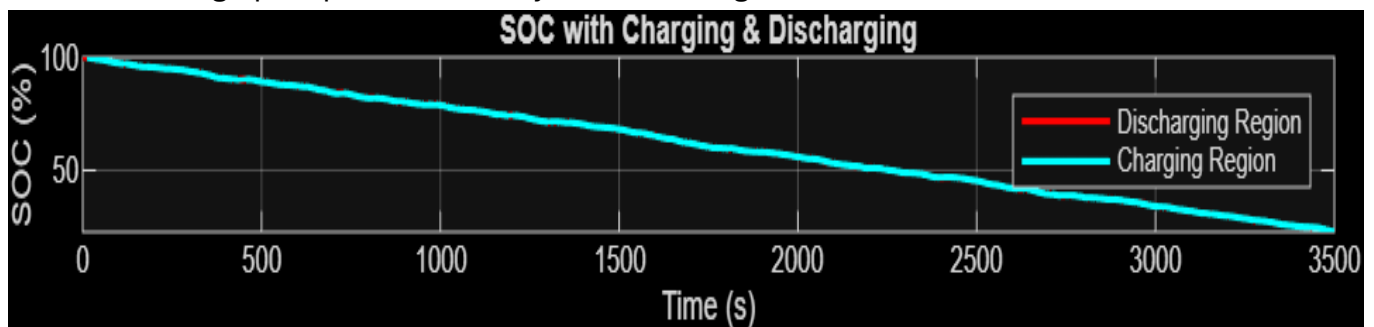
- SOC gradually decreases as distance increases
- Longer travel consumes more battery energy
- Regen braking slightly slows SOC depletion

Purpose:

- Study EV range behaviour
- Estimate battery utilization over distance

3. SOC vs Time

This graph represents battery State of Charge variation over time.

**Observation:**

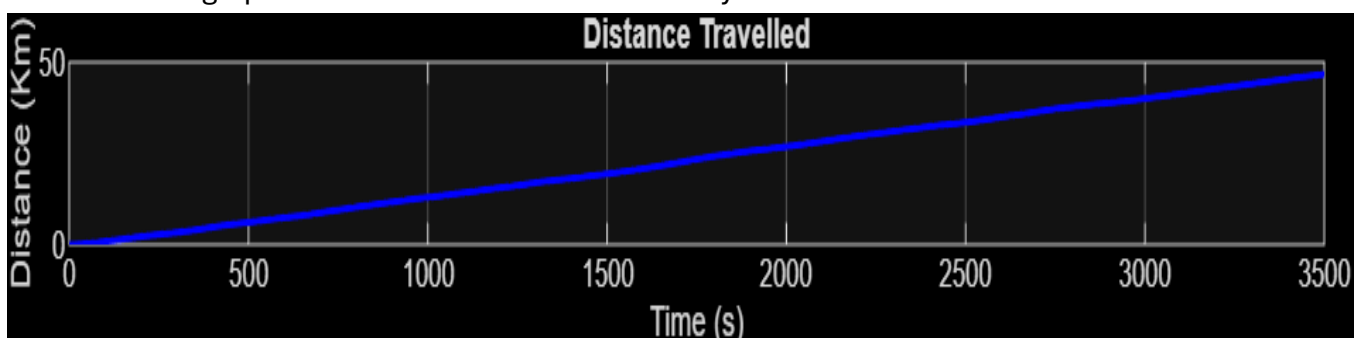
- SOC decreases during power consumption
- Small increases occur during regenerative braking
- Long simulation time shows discharge trend clearly

Purpose:

- Analyse battery depletion
- Observe energy recovery effect

4. Distance vs Time

This graph shows total distance travelled by the EV.



Observation:

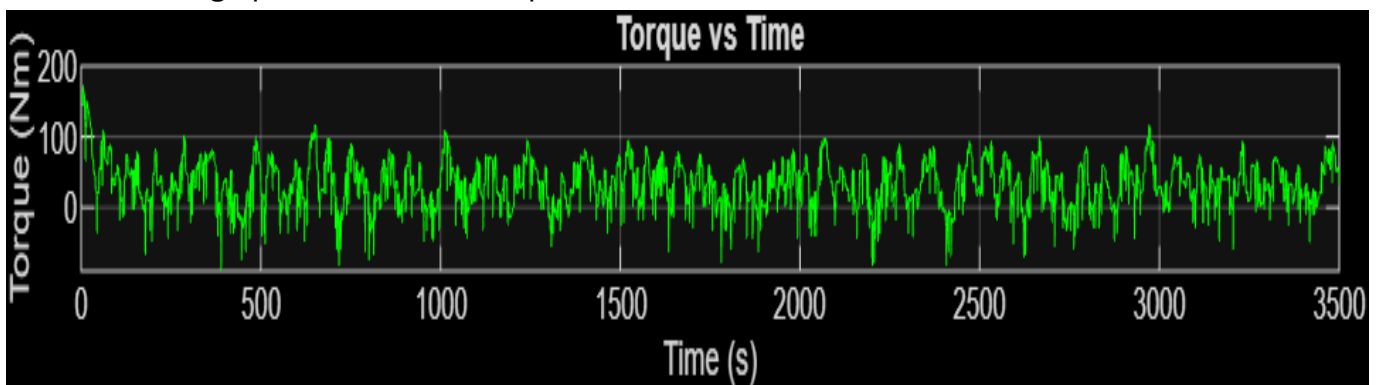
- Distance continuously increases
- Slope changes depending on vehicle speed
- Faster vehicle speed results in steeper curve

Purpose:

- Measure travelled distance
- Validate velocity integration

5. Torque vs Time

This graph shows motor torque demand.

**Observation:**

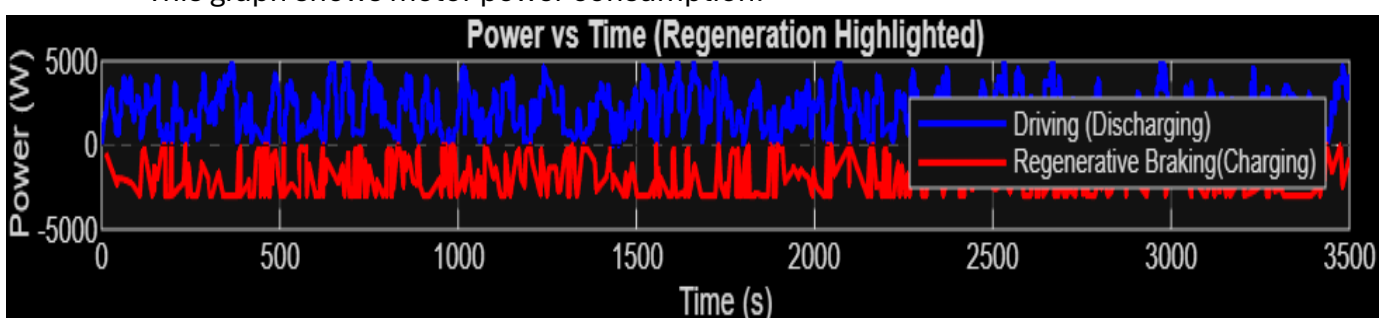
- High torque during acceleration
- Lower torque during cruising
- Negative torque may appear during regen braking

Purpose:

- Study motor load demand
- Understand drivetrain response

6. Power vs Time

This graph shows motor power consumption.



Observation:

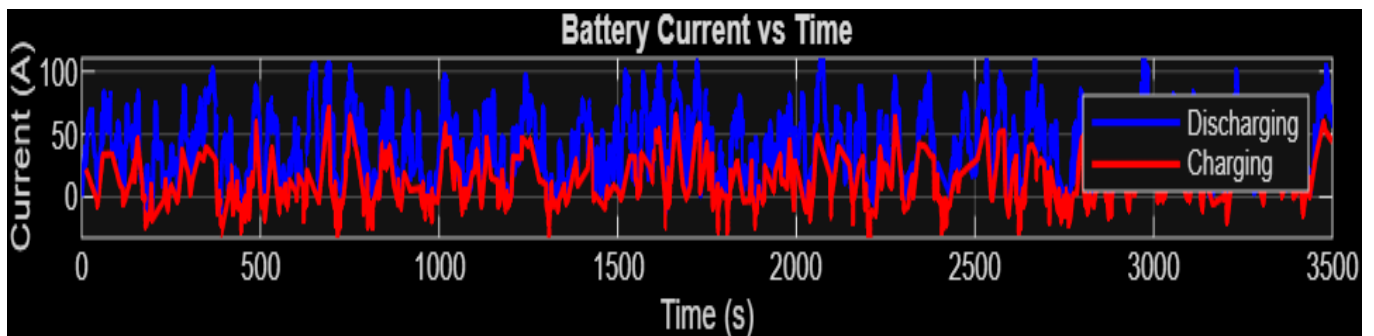
- Positive power indicates battery discharge
- Negative power indicates regenerative braking
- High power peaks occur during acceleration

Purpose:

- Analyse energy consumption
- Identify regen regions

7. Current vs Time

This graph shows battery current flow.



Observation:

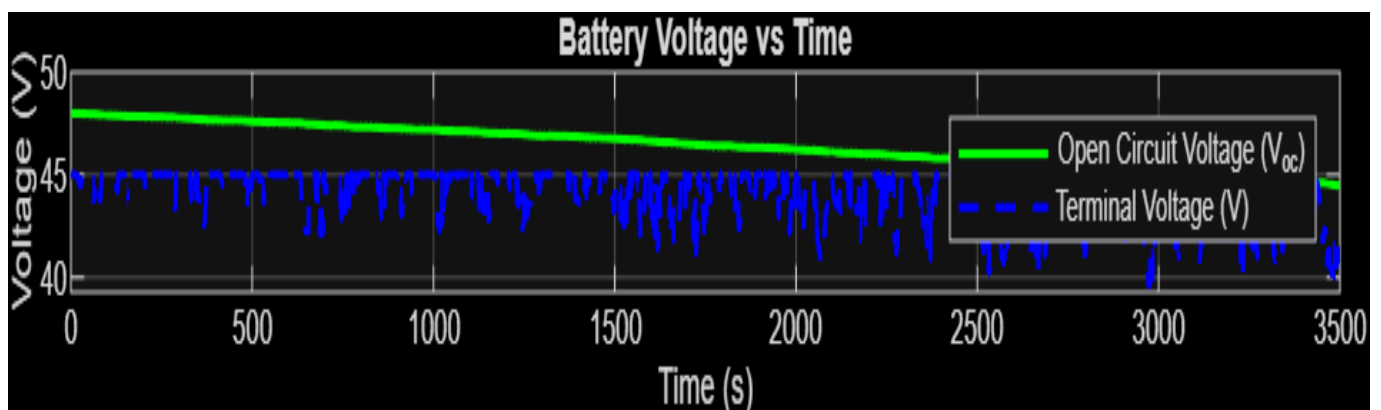
- Positive Current → Battery Discharge
- Negative current → Charging during Regeneration
- Current varies with motor load

Purpose:

- Analyse battery loading conditions
- Understand current demand

8. Voltage vs Time

This graph shows battery voltage variation.



Observation:

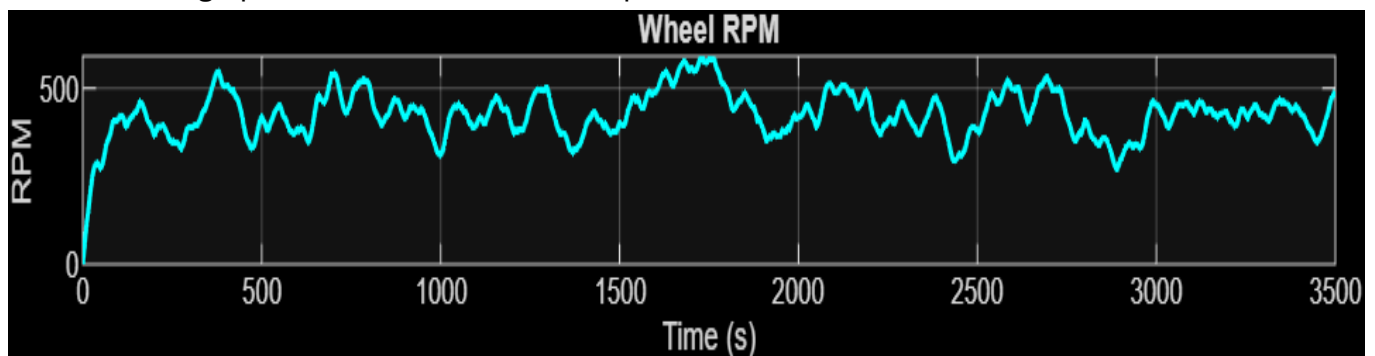
- Voltage decreases with SOC depletion
- Voltage may fluctuate with load demand
- Voltage stabilizes during steady driving

Purpose:

- Monitor battery operating condition
- Analyse voltage behaviour

9. RPM vs Time

This graph shows motor rotational speed.

**Observation:**

- RPM increases with vehicle speed
- RPM decreases during deceleration
- Closely follows velocity trend

Purpose:

- Analyse motor speed characteristics
- Study drivetrain performance

OVERALL RESULT SUMMARY

The simulation successfully demonstrates:

- Vehicle longitudinal dynamics
- Motor torque and RPM response
- Battery energy consumption
- SOC depletion behaviour
- Regenerative braking effect
- Electrical parameter variation
- EV performance over long-duration simulation

These results validate the EV powertrain model developed in MATLAB.

CONCLUSION

This project successfully develops and simulates an Electric Vehicle (EV) Powertrain model using MATLAB. The simulation demonstrates the relationship between vehicle dynamics, motor performance, battery energy consumption, and regenerative braking under dynamic driving conditions.

The model calculates important vehicle parameters such as speed, torque, power, current, voltage, distance travelled, RPM, and battery State of Charge (SOC). By considering resistive forces such as rolling resistance and aerodynamic drag, the simulation provides a realistic approximation of EV longitudinal motion.

The inclusion of regenerative braking improves the model by allowing partial energy recovery during braking conditions, which helps increase battery efficiency. The generated graphs provide clear visualization of vehicle behaviour and battery performance over time.

Overall, the project provides a strong understanding of EV powertrain operation, energy flow, and MATLAB-based numerical simulation. It serves as an effective educational model for studying electric vehicle systems and forms a foundation for more advanced EV simulations involving motor efficiency, battery modelling, drive cycles, and real-world operating conditions.

REFERENCES

1. Electric Vehicle Technology Explained — Used for understanding EV fundamentals, battery systems, and powertrain concepts.
2. MATLAB Documentation — Used for MATLAB syntax, numerical simulation, plotting, and array operations.
3. Fundamentals of Vehicle Dynamics — Reference for vehicle longitudinal dynamics and force analysis.
4. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles — Used for EV drivetrain and regenerative braking concepts.
5. Basic principles of:
 - Newton's Laws of Motion
 - Vehicle longitudinal dynamics
 - Battery SOC estimation
 - Power and torque relationships
 - Energy conservation concepts
6. Academic lecture notes and engineering textbooks related to:
 - Electric Vehicles
 - Vehicle Dynamics
 - Battery Modelling